



STEM SMART Single Maths 11 - Using Differentiation >

Stationary Points

A-level Maths Topic Summaries - Calculus

Subject & topics: Maths | Calculus | Differentiation Stage & difficulty: A Level P3

Fill in the blanks to complete the summary notes on stationary points.

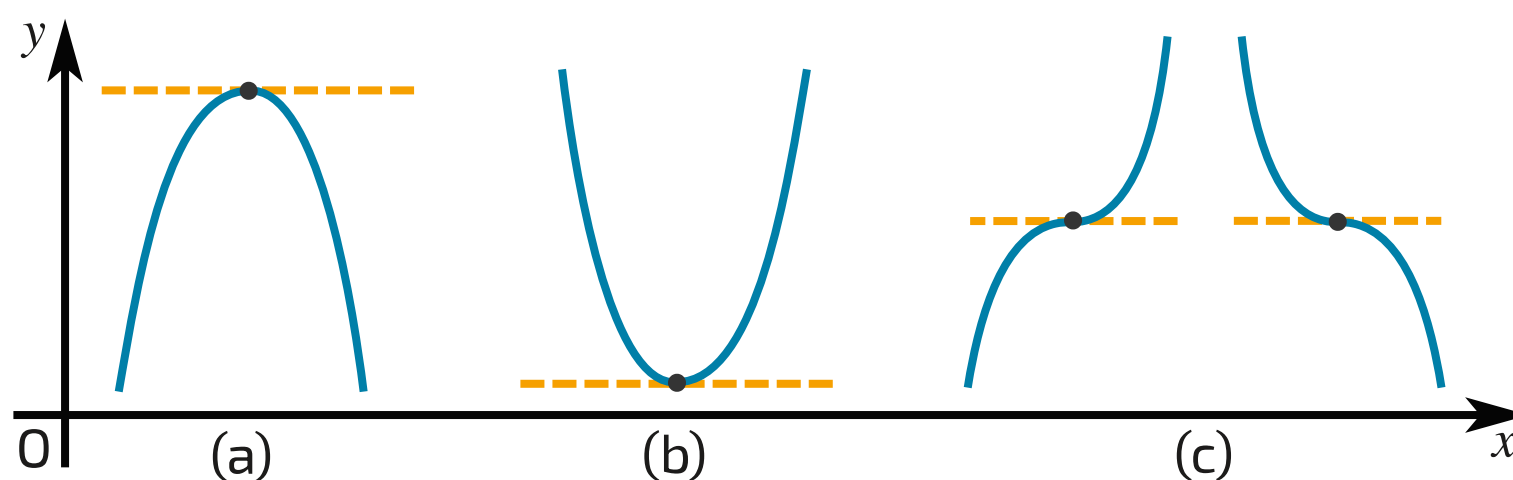


Figure 1: Stationary points: (a) a maximum (b) a minimum (c) a point of inflection.

At a stationary point, the tangent is . Therefore, at any stationary point

$$\frac{dy}{dx} \text{ }$$

There are three types of stationary point: maxima, minima and points of inflection.

At a maximum, $\frac{d^2y}{dx^2}$.

At a minimum, $\frac{d^2y}{dx^2}$.

If $\frac{d^2y}{dx^2} = 0$, a stationary point . To determine which we need to look at the values of the function, or the values of the gradient, on either side of the stationary point.

Items:

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Stationary Points 1

Pre-Uni Maths for Sciences J2.1

Subject & topics: Maths | Calculus | Differentiation Stage & difficulty: A Level P2

Find the coordinates, nature and number of the stationary points of the following functions.

Part A

$$y = 2x^3 - 24x - 5$$

$$y = 2x^3 - 24x - 5$$

Give the stationary points in order of increasing x -coordinate.

There is a at (,)

There is a at (,)

Part B

$$y = 2x^3 - 5x^2 + 4x + 6$$

$$y = 2x^3 - 5x^2 + 4x + 6$$

Give the stationary points in order of increasing x -coordinate. Enter fractions as improper fractions in their simplest form (e.g. 4/3).

There is a at (,)

There is a at (,)

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Question

Maxima and Minima: Problems 2ii

Subject & topics: Maths **Stage & difficulty:** A Level P2

A curve has equation $y = 3x^3 - 7x + \frac{2}{x}$

Part A

Verify stationary point

Fill in the blanks to verify that the curve $y = 3x^3 - 7x + \frac{2}{x}$ has a stationary point when $x = 1$.

At a stationary point the value of $\frac{dy}{dx}$ is .

For the curve, the first derivative is

$$\frac{dy}{dx} = \text{ } x^2 + \text{ } + \text{ } x^{-2}$$

Hence, when $x = 1$,

$$\frac{dy}{dx} = \text{ }$$

Therefore, the curve has a stationary point when $x = 1$.

Part B

Nature of stationary point

Fill in the blanks to determine the nature of the stationary point when $x = 1$.

The second derivative of $y = 3x^3 - 7x + \frac{2}{x}$ is:

$$\frac{d^2y}{dx^2} = \text{[] } x + \text{[] } x^{-3}$$

Hence, when $x = 1$,

$$\frac{d^2y}{dx^2} = \text{[]}$$

Therefore, the stationary point when $x = 1$ is (maximum / minimum / point of inflection):

Part C

Tangent to curve

The tangent to the curve at this stationary point meets the y -axis at the point Q. Find the y -coordinate of Q.

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Stationary Points 1ii

Subject & topics: Maths **Stage & difficulty:** A Level P2

The curve $y = x^3 - kx^2 + x - 3$ has two stationary points.

Part A

Differentiate

Find $\frac{dy}{dx}$.

The following symbols may be useful: k , x

Part B

Find k

Given that there is a stationary point when $x = 1$, find the value of k .

The following symbols may be useful: k

Part C

Differentiate twice

Find $\frac{d^2y}{dx^2}$.

The following symbols may be useful: x

Hence determine whether the stationary point is a minimum or a maximum.

☐ Minimum

☐ Maximum

Part D

Find coordinate

Find the x -coordinate of the other stationary point.

The following symbols may be useful: x

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Stationary Points 2ii

Subject & topics: Maths **Stage & difficulty:** A Level P2**Part A****Find coordinate**

Find the coordinates of the stationary points on the curve $y = x^3 - 3x^2 + 4$.

(,)

Part B**Stationary point**

Determine the natures of the stationary points of $y = x^3 - 3x^2 + 4$.

- ☐ The stationary point with the smallest x -coordinate is a minimum point and the stationary point with the largest x -coordinate is a maximum point.
- ☐ The stationary point with the smallest x -coordinate is a maximum point and the stationary point with the largest x -coordinate is a minimum point.
- ☐ There is a single inflection point

Part C

Range of x

For what range of values of x does $x^3 - 3x^2 + 4$ decrease as x increases?

Construct your answer from the items below.

Items:

- <
- >
- x
- $< x <$
- $\leq x \leq$
- $< x \text{ or } x <$
- $\leq x \text{ or } x \leq$
- $\leq$
- $\geq$
- 2
- 1
- 0
- 1
- 2
- 3
- 4

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Maxima and Minima: Problems 1ii

Subject & topics: Maths **Stage & difficulty:** A Level P2

Figure 1 shows a rectangular enclosure, with a wall forming one side. A rope, of length 20 metres, is used to form the remaining three sides. The width of the enclosure is x metres, and the area of the enclosure is $A \text{ m}^2$.



Figure 1: The rectangular enclosure.

Part A

Express as equation

Show that A can be expressed in the form $px - qx^2$, and find this expression.

The following symbols may be useful: A , x

Part B

Use differentiation

Use differentiation to find the maximum value of the area of the enclosure, $A \text{ m}^2$.

Enter your value of A :

The following symbols may be useful: A

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Maxima and Minima: Problems 1i

Subject & topics: Maths **Stage & difficulty:** A Level P2

A cuboid has a volume of exactly 8 m^3 . The base of the cuboid is a square with side length x metres. The surface area of the cuboid is $A \text{ m}^2$.

Part A

Find expression for A

Show that A can be expressed in the form $ax^2 + \frac{b}{x}$, where a and b are constants, and find this expression.

The following symbols may be useful: x

Part B

Find $\frac{dA}{dx}$

Find $\frac{dA}{dx}$.

The following symbols may be useful: x

Part C

Find minimum

Find the value of x which gives the smallest surface area of the cuboid.

The following symbols may be useful: x

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Differentiating Powers 7

Pre-Uni Maths for Sciences J2.3

Subject & topics: Maths | Calculus | Differentiation **Stage & difficulty:** A Level P2

A quadratic function has the form $y = a + bx + cx^2$ where a , b and c are constants. It has a stationary point at $(2, 2)$ and, at $x = 1$, the tangent to the curve has a gradient of -2 .

Find the values of a , b and c . (In practice, with the information given, you will need to find b and c before you can find a .)

$$y = \boxed{} + \boxed{} x + \boxed{} x^2$$

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Stationary Points 3

Pre-Uni Maths for Sciences J2.5

Subject & topics: Maths | Calculus | Differentiation **Stage & difficulty:** A Level C2

Part A

Maximum height of a projectile

A particle is fired upwards into the air with an initial speed w and moves subsequently under the influence of gravity with an acceleration g downwards, such that its height h at time t is given by $h = wt - \frac{1}{2}gt^2$, where w and g are constants. Find an expression for its maximum height above its initial position.

The following symbols may be useful: g , h , w

Part B**Potential energy of two molecules**

The potential energy of two molecules separated by a distance r is given by

$$U = U_0 \left(\left(\frac{a}{r} \right)^{12} - 2 \left(\frac{a}{r} \right)^6 \right)$$

where U_0 and a are positive constants. The equilibrium separation of the two molecules occurs when the potential energy is a minimum.

Find an expression for the equilibrium separation of the molecules.

The following symbols may be useful: U , U_0 , a , r

Find an expression for the potential energy when the molecules are at their equilibrium separation.

The following symbols may be useful: U , U_0 , a , r

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